Autonomous Retail Researcher Agent

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Academic Year:2025-26

Problem Statement & Objectives

1. Problem Statement
2. Project Objectives
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Proposed Solution

1. Key features *(Just mention key features here no need to go into details)*
2. Overall Architecture / Workflow
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Results & Output

Add the below details here:

1. *Screenshots / outputs*
2. *Reports / dashboards / models*
3. *Key outcomes*

Conclusion

Mention a brief conclusion about your project summing up everything you have worked on and the key learning.

Future Scope & Enhancements

Problem Statement & Objectives

1. Problem Statement

The retail industry is highly dynamic and competitive, requiring businesses to continuously monitor market trends, consumer preferences, product demand, and competitor activities. In today's digital era, a vast amount of retail-related information is available online across multiple platforms such as e-commerce websites, research articles, blogs, and industry reports. However, manually collecting, analyzing, and summarizing this information is a time-consuming and labor-intensive process. Retail analysts often spend several hours browsing different sources, verifying the authenticity of information, extracting relevant insights, and preparing structured reports.

This manual approach leads to several challenges, including information overload, inconsistent data quality, delayed decision-making, and increased operational costs. Furthermore, human researchers may overlook critical insights due to fatigue or limited processing capacity. Businesses require faster access to reliable information to make data-driven decisions, identify emerging trends, and maintain a competitive advantage in the market.

With the rapid advancement of Artificial Intelligence and Large Language Models (LLMs), it is now possible to automate many aspects of the research process. Agentic AI systems, which involve multiple autonomous agents working collaboratively, provide a promising solution for automating complex workflows such as web-based research and data analysis. These intelligent systems can perform tasks independently, coordinate with other agents, and generate meaningful outputs with minimal human intervention.

Therefore, the problem addressed in this project is the lack of an efficient, automated, and intelligent system capable of performing retail research autonomously. The need is to design and implement an **Autonomous Retail Researcher Agent** that can accept user queries, gather information from multiple reliable sources, analyze collected data, summarize key findings, and store structured reports for future use.

2. Project Objectives

The primary objective of this project is to design and develop an intelligent, LLM-powered autonomous researcher agent capable of performing retail research tasks efficiently and accurately. The project aims to automate the traditional research workflow by integrating advanced artificial intelligence techniques and multi-agent collaboration frameworks.

One of the key objectives is to implement a multi-agent architecture using CrewAI, where different agents are responsible for specific tasks such as information retrieval, analysis, summarization, and data storage. This modular design enhances system flexibility and improves performance by distributing tasks across specialized components.

Another important objective is to integrate web search tools that allow the system to collect real-time information from multiple online sources. By leveraging search tools, the agent can access up-to-date retail data, market trends, and competitor information. This ensures that the generated insights are relevant and reliable.

The project also aims to implement natural language processing capabilities using Large Language Models. These models are responsible for understanding user queries, extracting meaningful information from collected data, and generating concise summaries. This improves the clarity and usability of research outputs.

Additionally, the system is designed to store generated reports in a persistent knowledge repository. This enables users to retrieve previous research findings without repeating the entire process. The repository acts as a centralized storage system that improves long-term usability and supports knowledge reuse.

Overall, the project objectives focus on improving research efficiency, reducing manual effort, enhancing data accuracy, and enabling intelligent decision-making in the retail domain.

3. Scope of the Project

The scope of this project includes the design, development, and implementation of an autonomous research system specifically focused on the retail industry. The system is capable of handling various types of retail-related queries, including market trend analysis, product demand research, competitor analysis, and industry insights generation.

The project primarily focuses on automating the process of web-based research. It allows users to enter natural language queries related to retail topics, which are then processed by the system to retrieve relevant data from online sources. The collected information is analyzed using intelligent algorithms, summarized using Large Language Models, and stored in structured formats for future reference.

Another important aspect within the project scope is the implementation of a **multi-agent architecture**. Each agent is responsible for performing a specific task within the research workflow. This modular design allows the system to handle complex workflows efficiently and ensures scalability for future enhancements.

The project also includes the development of a **knowledge repository** for storing generated reports. This repository enables long-term storage of research outputs, allowing users to access previously generated insights without performing repetitive searches.

However, the project has certain limitations. The system relies on internet connectivity to retrieve online information, making it dependent on network availability. Additionally, the accuracy of generated insights depends on the reliability of data sources accessed during the research process. The current implementation focuses on text-based data analysis and does not include multimedia data processing such as images or videos.

Despite these limitations, the project scope provides a strong foundation for building advanced autonomous research systems capable of supporting real-world retail analytics and decision-making.

Proposed Solution

1. Key Features

- Autonomous retail research capability
- Multi-agent collaboration using CrewAI
- Integration with web search tools
- Natural language query processing
- Automated data collection
- Intelligent data analysis
- LLM-based summarization
- Persistent knowledge storage
- Automated report generation
- Modular system architecture
- Real-time information retrieval
- Scalable workflow management

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2. Overall Architecture / Workflow

The overall architecture of the Autonomous Retail Researcher Agent is designed to support efficient task execution through coordinated interactions between multiple agents. The workflow begins when a user enters a query related to retail research. This query is processed by the system and passed to the main controller agent.

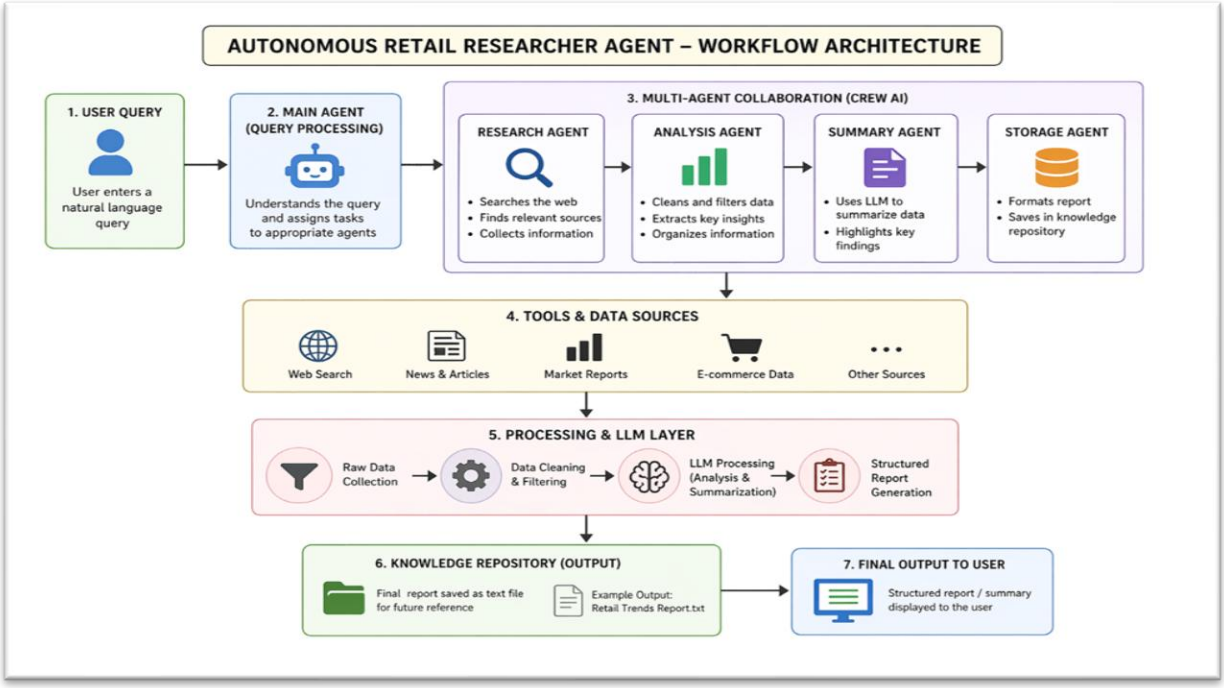
The main controller agent is responsible for understanding the query and assigning tasks to specialized agents within the CrewAI framework. The Research Agent retrieves relevant data from web-based sources using integrated search tools. This data may include articles, reports, blogs, and market statistics.

Once the data is collected, it is forwarded to the Analysis Agent, which processes and filters the information to remove irrelevant content. The processed data is then passed to the Summary Agent, where Large Language Models generate concise summaries of key findings. These summaries are structured in a readable format suitable for reporting.

Finally, the Storage Agent saves the generated report in the knowledge repository as a text file. The stored reports can be accessed later for reference or further analysis.

This architecture ensures efficient collaboration between agents and enables automated execution of complex research tasks.

Workflow Explanation



Tools and Technologies Used

1. Python Programming Language

Python was used as the primary programming language for developing the entire system. It is widely used in artificial intelligence and machine learning applications due to its simplicity, flexibility, and availability of powerful libraries. Python provides strong support for automation, data processing, and API integration, making it suitable for building agent-based systems.

In this project, Python was used to implement agent logic, workflow execution, data processing, and integration with external tools such as search APIs and Large Language Models. The language also enabled easy debugging and rapid development of system components. Python's extensive ecosystem of libraries helped in simplifying complex tasks such as natural language processing and web automation.

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2. LangChain Framework

LangChain is a powerful framework designed for building applications using Large Language Models. It provides tools for managing prompts, chaining tasks, and integrating external resources into AI workflows. In this project, LangChain was used to structure the flow of information between different stages of the research process.

LangChain allowed the system to connect language models with external data sources, enabling efficient information retrieval and processing. It also supported prompt engineering techniques that improved the accuracy of generated summaries. By using LangChain, the system achieved modular workflow design, which enhanced scalability and maintainability.

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3. CrewAI Framework

CrewAI was used to implement the multi-agent architecture of the system. It allows multiple agents to collaborate and perform tasks in a coordinated manner. Each agent in the system was assigned a specific responsibility, such as data collection, analysis, summarization, or storage.

CrewAI manages communication between agents and ensures that tasks are executed in the correct sequence. This multi-agent approach improves efficiency and allows the system to handle complex workflows more effectively. The framework also supports scalability, enabling additional agents to be added in future versions without major modifications to the system.

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4. Large Language Model (LLM API)

A Large Language Model (LLM) API was used to provide natural language understanding and summarization capabilities. The LLM plays a crucial role in interpreting user queries, analyzing collected data, and generating structured summaries.

The model processes large amounts of text and identifies important patterns, trends, and insights. It converts raw data into meaningful information that can be easily understood by users. The use of an LLM significantly improves the quality of generated outputs and ensures consistency in report formatting.

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5. Web Search Tools

Web search tools were integrated into the system to enable real-time data retrieval from online sources. These tools allow the research agent to gather updated information related to retail trends, market reports, and product analysis.

The search tools ensure that the system accesses reliable and relevant data from multiple sources. This improves the accuracy of insights generated by the system. Web search integration is essential for enabling dynamic and real-time research capabilities.

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6. Text-Based Knowledge Repository

A text-based knowledge repository was used to store generated research reports. This repository acts as a persistent storage system that saves structured summaries for future reference. Reports are stored in text files with meaningful names, making retrieval simple and efficient.

The repository enables knowledge reuse, allowing users to access previously generated insights without repeating the research process. This feature improves system efficiency and supports long-term data management.

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7. Prompt Engineering Techniques

Prompt engineering techniques were used to design structured prompts that guide the language model in generating accurate outputs. These prompts define the format, style, and content of the generated summaries.

By carefully designing prompts, the system ensures that outputs remain consistent and relevant to user queries. Prompt engineering also improves the quality of analysis performed by the LLM. This technique plays a significant role in enhancing system reliability and performance.

8. Operating System

The system was developed and tested on a Windows-based operating system. The operating system provides the environment required to execute Python scripts, manage dependencies, and run supporting tools.

The OS also supports file handling operations required for storing reports in the knowledge repository. It provides compatibility with development tools such as command-line interfaces and package managers.

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9. Integrated Development Environment (IDE)

An Integrated Development Environment (IDE) such as Visual Studio Code (VS Code) was used to write, debug, and test the program code. The IDE provides features such as syntax highlighting, debugging tools, and extension support.

Using an IDE improved development speed and reduced coding errors. It also allowed easy management of project files and libraries.

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10. Version Control System (Git & GitHub)

Git and GitHub were used for version control and project management. These tools helped track changes made to the project files and maintain different versions of the code.

GitHub was used to store the project repository, making it easier to share code and collaborate with team members. Version control ensures code safety and allows easy rollback to previous versions if required.

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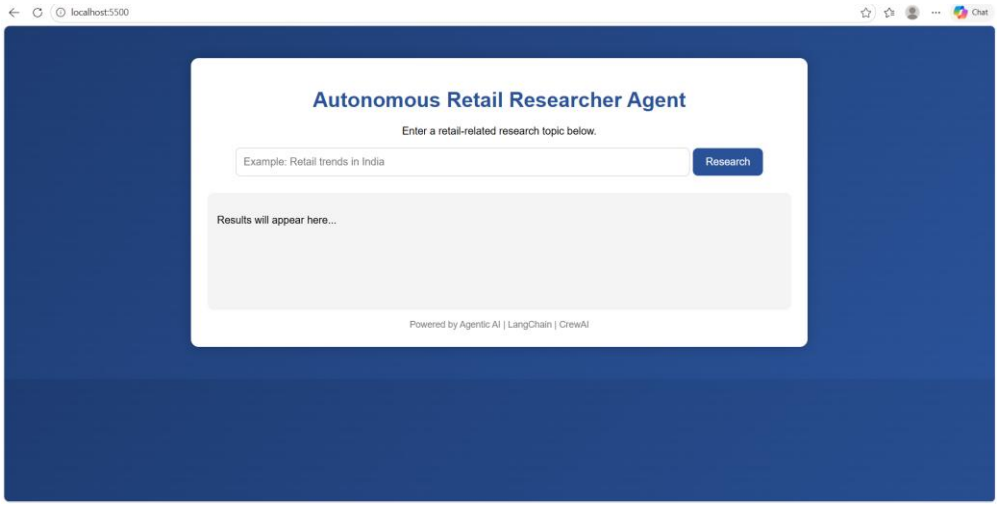
11. API Integration Tools

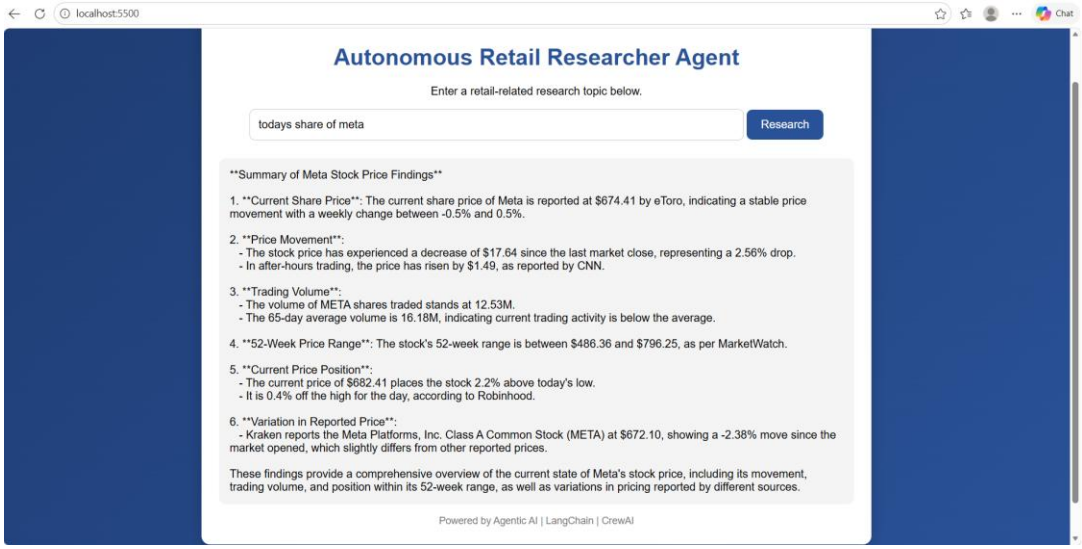
API integration tools were used to connect the system with external services such as language models and web search engines. These APIs allow seamless communication between the system and external platforms.

API integration ensures that the system can retrieve information and generate outputs efficiently. It also enables the system to extend its capabilities by connecting with additional services in the future.

Results & Output

1. ***Screenshots / outputs***





2. Reports / Dashboards / Models

The Autonomous Retail Researcher Agent generates structured reports based on user-provided research queries. The reports are designed to present complex retail-related data in a clear, concise, and organized format. The system processes the input query, retrieves relevant information from multiple sources, and produces a detailed summary containing key insights related to the requested topic.
Generated Reports
The system produces text-based analytical reports that include important retail insights such as market trends, stock performance, product demand, and competitor analysis. Each report is automatically structured into logical sections to improve readability and usability.
Based on the screenshot provided, the generated report titled "Summary of Meta Stock Price Findings" includes multiple informative sections such as:

- Current Share Price
- Price Movement
- Trading Volume
- 52-Week Price Range
- Current Price Position
- Variation in Reported Price
- Overall Summary of Findings

These structured sections allow users to quickly understand the current status of a retail-related entity such as stock performance or market conditions. The reports are displayed directly in the user interface and can also be stored in a knowledge repository for future use.
The reports generated by the system are designed to support decision-making processes by providing summarized insights instead of lengthy raw data. This reduces manual analysis time and ensures that users receive actionable information quickly.

Dashboards

The project currently includes a web-based user interface dashboard developed using frontend technologies that allow users to interact with the system easily. The dashboard includes:

- A search input field where users can enter retail-related queries
- A "Research" button that triggers the research workflow
- A results display panel that shows generated summaries
- A clean and user-friendly layout for better readability

The dashboard displays outputs in a structured format, allowing users to view research findings instantly after submitting a query. It also provides visual clarity through organized text sections and formatted bullet points.
Although the current implementation uses a text-based output panel, the system architecture supports future integration of graphical dashboards using visualization tools such as charts and trend graphs. This would enhance the visual representation of insights and improve overall user experience.

Models Used

The system integrates multiple models and frameworks to enable intelligent automation and research capabilities. The primary models used in the project include:
Large Language Model (LLM)
A Large Language Model is used to interpret user queries, analyze retrieved data, and generate meaningful summaries. The model processes large volumes of textual information and extracts key insights relevant to the query.
Functions performed by the LLM include:

- Understanding natural language queries
- Identifying relevant information
- Summarizing collected data
- Generating structured research reports

This model is the core intelligence component of the system, enabling it to perform advanced text analysis and produce high-quality outputs.

LangChain Framework

LangChain is used to manage the interaction between the Large Language Model and external tools. It enables the chaining of tasks, allowing the system to move data from one stage to another efficiently.
Functions performed by LangChain include:

- Managing prompts
- Connecting tools to language models
- Controlling workflow execution
- Handling response generation

LangChain improves the modularity and flexibility of the system architecture.

CrewAI Multi-Agent Model

CrewAI is used to implement the multi-agent system architecture. Multiple agents collaborate to complete different stages of the research workflow.
Agents implemented in the system include:

- Research Agent – Collects relevant data from web sources
- Analysis Agent – Filters and processes retrieved data
- Summary Agent – Generates structured summaries
- Storage Agent – Saves results into the knowledge repository

This multi-agent model ensures efficient task distribution and improves overall system performance.

3. Key Outcomes

The development and implementation of the Autonomous Retail Researcher Agent resulted in several significant outcomes that demonstrate the effectiveness and reliability of the system. The project successfully achieved its objective of automating retail research tasks using intelligent agent-based technologies.
One of the major outcomes of the project is the successful development of an **autonomous research system** capable of retrieving and analyzing retail-related information without manual intervention. The system demonstrated the ability to process natural language queries

and generate structured outputs in real time. This significantly reduced the time required to perform research tasks.

Another important outcome is the successful implementation of a **multi-agent architecture** using CrewAI. The collaboration between multiple agents allowed the system to perform complex workflows efficiently. Each agent performed its assigned task independently, ensuring smooth execution of the overall process.

The project also achieved **accurate and structured report generation**, as shown in the output screenshot. The generated reports contained well-organized information such as share prices, price movements, trading volumes, and statistical ranges. This structured output improves readability and supports quick understanding of research findings.

Additionally, the system demonstrated **real-time information retrieval capability**, enabling users to access updated data from multiple sources. This ensures that the generated insights remain relevant and reliable.

Another significant outcome is the creation of a **persistent knowledge repository**, where generated reports can be stored and retrieved later. This feature enhances knowledge management and supports long-term usability of the system.

Overall, the key outcomes of the project include:

- Successful automation of retail research workflows
- Implementation of multi-agent collaboration
- Real-time data retrieval from online sources
- Generation of structured and meaningful reports
- Reduction in manual research effort
- Improved efficiency and accuracy
- Development of a scalable and modular system
- Enhanced user interaction through a web-based interface

These outcomes demonstrate the practical applicability of Agentic AI technologies in solving real-world research challenges and highlight the potential of autonomous systems in modern industries.

Conclusion

The **Autonomous Retail Researcher Agent** project successfully demonstrates the practical implementation of Agentic AI in automating retail research tasks. The system integrates Python, LangChain, CrewAI, and Large Language Models to create an intelligent multi-agent environment capable of processing user queries, collecting data from web sources, analyzing information, and generating structured summaries automatically.

The project achieved its primary objective of reducing manual research effort by enabling automated information retrieval and summarization. The multi-agent architecture improved workflow efficiency by assigning specific responsibilities to different agents such as research, analysis, summarization, and storage. Additionally, the system provides a user-friendly interface that allows users to interact with the agent using natural language queries.

Overall, this project highlights the effectiveness of autonomous AI-driven systems in handling complex research workflows. It demonstrates how intelligent agents can improve productivity, accuracy, and decision-making in the retail domain. The knowledge gained during this project includes multi-agent system design, workflow automation, API integration, and intelligent data processing, which are valuable skills for developing advanced AI-based applications.

Future Scope & Enhancements

The Autonomous Retail Researcher Agent can be further enhanced by integrating additional features that improve its performance, usability, and scalability. One possible future improvement is the integration of **real-time data visualization dashboards**, allowing users to view insights through charts and graphs instead of only text-based reports.

Another important enhancement is the use of **database systems** such as MySQL or MongoDB to replace text-based storage. This would improve data organization, retrieval speed, and scalability for handling large datasets. The system can also be expanded to support **voice-based query input**, making it more user-friendly and accessible.

In the future, the project can be extended to other domains such as healthcare, finance, and logistics, demonstrating the versatility of autonomous research agents. Deploying the system on cloud platforms such as AWS with Docker containerization can also improve reliability and allow access from anywhere. These enhancements would transform the current system into a more advanced, scalable, and industry-ready autonomous research platform.